ARGUMENTS: DEDUCTIVE AND INDUCTIVE REASONING

INDUCTIVE AND DEDUCTIVE LOGIC

INTRODUCTION

This lesson concerns logic. Logic is described as the descriptive of discourse. Discourse in its turn is connected thought. Why study logic? The purpose to this concern lies simply in the realization that, wherever people debate, discuss or argue, logic is usually a sort of court of appeal in the background; whenever, we debate matters in our own minds, a silent logic usually arbitrates. No one in his or her senses will willingly and persistently defy a clear verdict by logic. Therefore, logic trains the mind to draw the right conclusion, and to avoid the wrong conclusion by formulating rules of inference to govern and guide debate (or reason) and to promote discovery.

Logic is defined at this stage as the study of reasoning. This is distinguishable from the way a psychologist, biologist, psychiatrist, sociologist etc could study reasoning. Reasoning in logic concerns the process in which we are able to move from given fact(s) or statement(s) to draw another fact or statement. For example, given that:

All human beings are mortal and that all Kenyans

are human beings, we can draw the conclusion

that All Kenyans are mortal.

Logic therefore is a study that investigates the connections (or lack of them) between statements.

But logicians do not study statements merely put together haphazardly or arbitrarily. Instead they concern themselves with statements joined together in a structured form. Such structured forms in which statements are put together are referred to as ARGUMENTS.

It is worth noting that the word "argument" has a broader usage in ordinary English than in logic. For example, we normally say of two people screaming abuses at each other in a local bar are having an argument. Again we say that one "is advancing an 'argument' in favor of position". In this usage, we normally tend to understand "argument" as referring to the reasons one is giving for holding the given position.

But in logic, an argument includes the reasons and the conclusion. So we can define an argument in logic as:

A group of statements, such that one of them is

considered to be the conclusion. The conclusion

is statement intended to follow from the other

statements which are given as reasons (or premises)

for it.

This definition, however, requires us to distinguish statements from other types of sentences. Ordinary language is made up of grammatically well-formed sentences. These can be:

Questions e.g. "what is you name?"

Wishes e.g. "May God bless you abundantly"

Exclamations e.g. "Wow, what a surprise!

Commands e.g. "Shut up"

Statements e.g. It is hot today"

What distinguishes statements from the rest of the sentences is that they are capable of being True or False. We cannot say so of the question,"what is your name?' that it is True or False. But we can say of the statement, "It is hot today" that it is True, or False, depending on when and where it is uttered. We therefore can say that only statements have Truth-value and all the other sentences do not have Truth-value. The Truth-value of a true statement is True and that of a false one is False.

NB All statements are sentences, but not all sentences are statements.

STANDARD FORM OF ARGUMENTS

In logic, arguments are presented in a certain standard form. This form requires that we state the premises (or reasons) first and then the first argument you encountered at the beginning of the lesson would be presented thus:

All human beings are mortal

All Kenyans are human beings.

Therefore, all Kenyans are mortal.

ARGUMENTS IN ORDINARY LANGUAGE

In ordinary speech or writing we normally do not follow the prescribed standard form of stating arguments above. Instead, we realize that our conclusions might appear at the beginning or in the middle of the argument. However, there are particular words which are employed in ordinary language that are referred to as ARGUMENT INDICATORS. Some of these are precisely premise-indicators and conclusion-indicators, which tell us what the premises and the conclusion are respectively.
For example:

"Peter is not a boy scout, since Peter does

not know how to tie a square knot and all boy scouts

know how to tie square knots'.

This can be put in standard logical form thus:

Peter does not know how to tie a square knot

All boy scouts know how to tie square knots

Therefore, Peter is not a boy scout.

The following is a list of some common premise-indicators:

Since	follows from
As	because
In as much as	may be inferred from
For this reason	is based on
For	one Two
First	Second

The following are some common conclusion-indicators:

Therefore	so
Hence	which shows that
Thus	We can derive that
It follows that	Consequently

NOTE: In identifying the premises and conclusion of an argument, we first look for our ordinary language argument indicators.

Why do we construct arguments?

We normally construct arguments for these reasons;

- To explain
- To predict
- To convince and persuade

This means therefore, that besides the premises-indicators, an examination of the context in which the argument is constructed and the motives behind its presentation can usually be helpful in interpreting arguments.

DEDUCTIVE AND INDUCTIVE REASONING

INTRODUCTION

When we arrange various arguments into the standard logical form, we realize that there are typically two types of logical reasoning that we normally engage in. Hence logic is conventionally divided into deductive logic and inductive logic.
An argument is considered to be deductive when the premises present absolutely conclusive evidence for the conclusion. Thus, a deductive argument can either be valid or invalid.

NOTE: An argument is deductively valid if and only if its underlying structure or form guarantees that: if its premises are all true, then its conclusion is also true.

This means that in deductively valid argument, when its premises are true then, it is IMPOSSIBLE that its conclusion is false. Otherwise, we involve ourselves in a contradiction.

But an argument is deductively invalid if and only if its underlying structure does not guarantee that; if its premises are all true, then so is its conclusion.

Notice that the above definitions of deductive validity or invalidity are not about the truth of the premises but rather the structure of the arguments. It is precisely the structure of the argument that guarantees that provided its premises are all true, then its conclusion will not be false.

Let us take an example to illustrate our point:

> All Senegalese are Africans.
>
> President Moi is a Senegalese
>
> Therefore, President Moi is an African.

We observe that the second premise in the argument above, is false because President Moi is not a Senegalese. Nevertheless, we are claiming that if these premises were all true, then the conclusion would also be true. Therefore, the above argument is a deductively invalid argument.

In pure logic, the truth is not the major concern. Validity instead is the major concern. However, as critical and creative thinkers we are constantly concerned about whether an argument has true premises.

Whenever an argument has all true premises and deductive validity, we say that it is a sound argument. Otherwise it is unsound.

NOTE: An argument is SOUND if and only if it is both deductively valid and has all true premises.

The reason why it is impossible for the conclusion of a deductively valid argument to be false if the premises are true is that the conclusion makes no factual claim that is not

implicitly made by the premises. What the conclusion does is merely to lay bare (or explicit) what was initially hidden (or implicit) in the premises. So, the conclusion actually does not venture to say anything more than what the premises contain.

It is noticeable therefore that deductive logic regards argument correctness as all or nothing, i.e. either it is valid or it is invalid. Inductive logic on the other hand, is concerned with connections from premises to concussions that allow for graduations. We can hence have a more or less strong inductive argument.

NOTE: An argument is inductively strong (or weak) in direct proportion to the likelihood of its conclusion being true on the assumption that its premises are true.

This means that if the premises of an inductive argument are true then it is IMPROBABLE that its conclusion is false. The degree of inductive strength depends entirely on how improbable it is that the conclusion is false given that the premises are true. Therefore, inductive arguments can be of varying degrees of strength ranging from very strong inductive arguments to weak inductive arguments.

The conclusion of an inductive argument asserts more than the premises. It therefore makes factual claims which lie beyond what the premises claim. A denial of the conclusion of an inductive argument does not lead us to a contradiction since we can have descriptive situations in which the premises remain true and yet the conclusion is false.

A typical example of inductive arguments are the weather forecasts of the meteorological department. On the basis of the data and information gathered during the day or throughout the week, they forecast what the whether is likely to be the following day. The forecast certainly ventures beyond what the premises contain because it refers to a day which has not yet arrived.

So in evaluating inductive arguments we should try to determine the probability or likelihood of the conclusion being true if the premises are all true. If the conclusion is likely to be true if the premises are true then we consider the argument to be inductively strong. But if the conclusion is only slightly supported by the premises, then we consider the argument to be inductively weak.

NOTE:

a) Arguments that are deductively invalid can possibly be inductively strong or inductively weak, depending on the strength of connection between the premises and the conclusion.

b) Arguments that are studied by inductive logic usually proceed from what has been experienced to what has not been experienced.

CONCLUSION

For analytical purposes, we have divided arguments into two categories namely, deductive and inductive. But this poses the danger of understanding the two types as separate and mutually exclusive.

Instead we should understand the two types of arguments as constituting a single, continuous scale of varying degrees of strength of the link between the premises and the conclusion.

We can diagrammatically represent this, relationship thus:

Arguments

Deductively valid

Degrees or inductive strength

Worthless

From the diagram we notice that deductively valid arguments have the strongest possible link (of necessity) between the premises and the conclusion. This is followed by the varying degrees of strength in inductive arguments with the most worthless having the weakest link (if any) between the premises and the conclusion.

At this juncture, we can therefore formulate a more technical definition of logic

Logic is the study of the strength of the Evidential link between the premises

And conclusions of arguments.

THE GENERAL AND THE SPECIFIC

At this juncture, I would like to draw your attention to one of the most widespread and misleading conceptions in logic. It is the simplistic contention that the distinction between deductive and inductive arguments depends on whether they proceed from general or specific statements.

According to this view, a deductive argument is defined as an argument that proceeds from general premises to specific conclusion so if we take a valid deductive argument like

All men are mortal

Socrates is a man

Socrates is mortal

We notice that the first premise makes a general claim about all men and the conclusion makes a claim about a specific man, namely, Socrates.

On the other hand, an inductive argument is defined as one that proceeds from the specific premise to the general conclusion. For example:

Water from Lake Victoria boiled at 100^0C

Water from Lake Turkana " "

Water from the rainfall " "

Water from Indian Ocean " "

Water from River Nile " "

Therefore, water boils at 100^0C.

We notice that the premises make claims about water from specific sources and the conclusion makes a claim about water in general, irrespective of its source. These examples can easily lend credence to this misconception about deductive and inductive arguments. In the previous section we did demonstrate that deductive and inductive arguments are not two types (except for analytical purposes). Instead, they constitute a continuous range of degrees of strength of link between the premises and the conclusion.

Perhaps some further examples can be employed here to demonstrate that we do have arguments which are deductively valid and yet they process either from general to general, or from specific to specific or even from specific to general.

1. From general to general

   All human beings are mortal

   All Kenyans are human beings

   Therefore, all Kenyans are mortal

2. From specific to specific

   The reader of this module is a student

   This module is for students of critical Thinking

   Therefore, the reader of this module is a student of critical Thinking

3. From specific to general

   One is a lucky number

   Three is a lucky number

   Five is a lucky number

   Seven is a lucky number

   Nine is a lucky number

Therefore, all odd numbers between 0 and 10 are lucky numbers

Likewise, we can get examples which illustrate that inductive arguments can proceed also from general to specific, general to general; specific to specific.

1. From general to specific

   All emeralds previously found have been green.

   The next emerald to be found will be green.

2. From general to general

   All students in this class are highly intelligent.

   All students in this class are strongly motivated to do well.

   No student in this class has a heavy workload.

   No student in this class has psychological difficulties that would interfere with his course work.

   All students in this class will do well.

3. From specific to specific

   Hitler was a dictator and was ruthless

   Stalin was a dictator and was ruthless

   Castro is a dictator

   Therefore, Castro is probably ruthless

FALLACIES: ERRORS IN REASONING

FALLACIES

INTRODUCTION

In this lesson, we are essentially going to discuss reasoning as a process. This discussion will be proceeded by an examination of what constitutes good and bad reasoning proper and improper reasoning or better still, correct and incorrect reasoning. In this regard, the lesson is going to provide an exposition and explication of the logical notion referred to as "Fallacy" or "Fallacies" in plural. This will be done with the objective of showing sources of error in reasoning and the imperatives for good reasoning. The lesson is thus going to provide the fundamentals for reasoning that entails inference(s) that is/are logically warranted. In other words, the lesson will ultimately present the gist for ensuring that conclusions arrived at in reasoning are properly or accurately drawn. Here, the ideal of good or accurate reasoning is going to be postulated.

In this lesson we have discussed how reasoning proceeds, the goal of reasoning, and how this goal may be achieved on one hand or fail to be achieved on the other. In this light, we have shown particular kinds and instances of error in reasoning, the sources of such errors, and how they can be avoided. These errors are what have been referred to as 'Fallacies'. However, the notion 'Fallacy' has been discussed in this lesson with a very specific context, logic. We have shown that we can analyze an argument or a piece of reasoning, and establish whether it is likely to be correct or not. It is also important to note hitherto that although one does not need to know expressly the rules and principles of correct reasoning (as conceived in logic), for one to argue and even argue well, knowledge of these rules enhances the ability of one to identify and avoid mistakes in reasoning.

Definition and Nature of fallacies

Richar Popkin and Arrum Stroll (1975; 248) define a fallacy as any sort of mistake in reasoning or inference; it is a term used to denote anything that causes an argument to go wrong. However, logicians use the word, not to designate any mistaken idea or false belief, but rather typical errors, mistakes that emanate commonly in ordinary discourse, and that render unsound the arguments in which they appear. It is important to note that for an argument to be fallacious, it must be such that it can easily be thought of as good (valid or cognent). In other words, the argument has to be psychologically persuasive notwithstanding the concealed structural or technical error or error with regard to the actual content of the argument for the argument to pass as fallacious. This is the sort of argument which, from the face of it, appears valid or cogent (good, proper, accurate, correct).

Any argument regardless of its subject or sphere is generally constructed for the purpose of providing that its conclusion is true. This is in line with the essence of reasoning which is that on the basis of certain proposition(s), certain other propositions or a certain proposition follows either of logical necessity or probability. That the truth of the

anteceding proposition(s) either guarantees that of the consequent propositions(s) or that such truth logically justifies, warrant or reasonably grounds that of the latter. The first category of proposition(s) is what is called the premise(s) of the argument the second or subsequent category of proposition(s) is what is referred to as the conclusion. The first category thus provides or at least is believed to provide evidence, grounds justification or warrant for the second category (conclusion). This step-by-step process from premise(s) to conclusion(s) is called inference. In inference, one starts with one or more propositions which have been accepted (or at least assumed accepted); then those propositions are used to arrive at a new proposition. If the inference is accurate (valid or logically probable) that proposition should also be accepted. That proposition can be used for inference later on. This, initially one can only infer things from the premises of the argument but as the argument proceeds, the number of statements or propositions available for inference increases.

In the attempt to prove that its conclusion is true, any argument can fail to fulfill this endeavor either by assuming a false proposition as one of its premises or by its premises not implying its conclusion i.e., there not being a logical relation between the premises and the conclusion. Regarding the first possibility, it is imperative to appreciate that every argument involves the claim that the truth of its conclusion follows from, or is implied, by the truth of its premises. However, the strength of a logician does not lie in the ability to establish the truth of propositions because the competent and reliable authority to establish such truth is the respective or relevant scholar who may be a sociologist, chemist etc. The competence, authority and strength of a logician rests in the ability to evaluate the logical relations that exist or at least claimed to exist between or among propositions. It is in this latter or second context that logicians lack of fallacies.

There are, broadly speaking two kinds of fallacies; formal and informal fallacies. The bottom-line of the distinction between formal and informal fallacies. The bottom-line of the distinction between formal and informal fallacies lies in the method of detecting the fallacy. If it is identifiable through mere inspection of the form of the argument, then it is formal. If, however, it is only identifiable through analysis of the content of the argument, then it is informal. However, the fact that an argument is fallacious does not at all mean that its conclusion is false.

FORMAL FALLACIES

Formal fallacies arise from error or defect in the structure or form of an argument. They are fallacies identified on the basis of the evaluation of the form or structure of an argument i.e. arise from formal error in argument. A formal fallacy is therefore to a great extent an error in deductive reasoning. In this case, the conclusion asserted does not follow of logical necessity such that despite the premises being true, the truth of the conclusion is not guaranteed from the logical view-point. This is a case of an invalid instance of reasoning, an invalid argument. In other words, when an argument which has the form of a syllogism seems valid, but is not, it is said to be fallacious. A syllogism is a two premised argument. There are five rules for determining the validity/invalidity of a

syllogistic argument and a violation of any of the five rules results into committing a fallacy. These rules are as follows:

Rule (1) In any valid categorical syllogism the middle term must be distributed at least once.

No priest is corrupt

Some priests are policemen

-Some policemen are not corrupt.

In the above example, the term 'priest' acts as a middle term in that by virtue of it a relationship or logical connection is established between the major and minor terms, in this case 'corrupt' people and 'policemen'. On the basis of an established link between the minor and major terms by the asserted relationship or link between the major terms the middle term and the minor term and the middle term, the implied relationship between the major and minor terms can be asserted in the conclusion. On other words, when reference is not made to all the members of the middle term either in the inclusive or exclusive sense, it becomes logically impossible to ascertain how the major and minor terms related to each other which means that inference becomes logically impossible. A violation of this rule results to the fallacy of the undistributed middle term or the fallacy of the undistributed middle.

e.g. All human beings are clever beings. All dogs are clever beings

All dogs are human beings.

In the above example, the middle term 'clever beings' is not distributed. Though in the first premise it is clear that all human beings are clever beings, it does not follow that all clever beings are human beings, reference is made to only a part of the class of clever beings who are human beings and that is all the same way, although it is clear that all dogs are clever beings,, it does not follow that all clever beings are dogs, in this circumstances the relationship between dogs and human beings is not certainly implied which make any conclusion with regard to how the two terms relate to each other logically impossible; thus rendering any conclusion uncertain. This makes the argument deductively invalid.

Rule 2. In any valid categorical syllogism, no term may increase its distribution i.e. no term should move from being undistributed in the premises to being distributed in the conclusion.

The rationale of this rule is that one cannot validly argue moving from some members of a class to all members of the class. Rather valid movement may be from all to some i.e. that since all are, some are. When this rule is violated the corresponding fallacy is the illicit process of the major term or minor term depending on the affected or relevant term which has a double status with regard to its distribution in the same argument.

e.g. All trees are plants

No vegetables are trees

No vegetables are plants

_____

Illicit process

Of the major

Term. INVALID

In the above example, 'plants' is the major term. It is undistributed in the first premise but it is distributed in the conclusion. This is invalid.

Rule 3. No valid categorically syllogism can have two negative premises. This is because negative premises are by nature exclusionary such that they deny class inclusion (separate minor from major term) either wholly or partially and that is all. This means that there is no link that is stated of the major and minor premises since all that is asserted is the exclusion of one class from another either in part or in whole. Concluding from the premises becomes logically impossible.

e.g.

No cats are dogs

No cows are dogs

Some cows are cats

_____

INVALID. Fallacy of Exclusive premises.

The corresponding fallacy is the fallacy of exclusion premises.

Rule 4.

In valid categorical syllogism, if a premise is negative, the conclusion must be negative. This is because a negative premise asserts and implies class exclusion such whatever conclusion is drawn must assert class exclusion also in part or in whole, as the case may be.

e.g.

All head-hunters are primitives

Some louderners are head-hunters

Some loudeners are head-hunters.

The above example is INVALID and commits the fallacy of drawing an affirmative Conclusion from negative premises.

Rule 5

Any valid categorical syllogism with a particular conclusion cannot have two universal premises. This rule however has limitations. There are two ways of interpreting categorical propositions, hypothetical and existential. The rule only holds when the interpretation of the relevant propositions is hypothetical not existential since in the latter case the argument is valid.

e.g. All P are M

No S are M

Some S are not P

_____

INVALID in the hypothetical interpretation, though VALID in the existential interpretation

The corresponding fallacy that results from a violation of this rule is the Existential fallacy.

Other examples of fallacies that are identifiable by the evaluation of the form or structure of an argument include:

1. The Fallacy of Affirming the consequent.
2. The Fallacy of denying the antecedent

E.g. (a) If P then Q } INVALID. Fallacy of affirming

Q } the consequent

Therefore, P }

If one contracts AIDS, then he dies John has died

Therefore, John had contracted AIDS

The above form of argument is fallacious because death can result from causes or anticidents other than just contracting AIDS. AIDS is only a sufficient condition for death but not a necessary condition for death. The above argument confuses sufficient condition for necessary condition. The argument can very easily deceive one to be valid because it appears or seems to take the form of a valid argument.

1, P > Q

2, P

_____

: Q

Modules Ponens (MP) is a valid argument form which proceeds by an affirmation of an auticedent warranting thereby an affirmation of the corresponding consequent. In other words, affirming the antecedent warrants affirming the consequent does not warrant affirming the antecedent. The consequent can only be denied on the basis of which the antecedent would be validly denied. This latter argument form of denying the consequent and on such basis denying the antecedent is called modus Tollens (MT) and it is a valid argument form as shown bellow:

1.If P, then q

3. not-q / not-p

If P then q

Not P

Therefore, not Q

_____

INVALID. Fallacy of denying the antecedent

If one contracts AIDS, then he will die.

John does not have AIDS

Therefore John will not die.

The rationale for the fallacy is that death can result from many causes or antecedents such that by one not having AIDS, one is still subject to death anyway. AID is just one of the causes for death i.e. the antecedent in the conditional is just one of the possible 'causes' of the consequent or one of the factors which precede the consequent, it is just sufficient condition for the consequent, it does not need to hold for the consequent to be observed. On this basis, by the move fact that one does not have AIDS as in the above example, there is no guarantee for immortality. The argument easily deceives because it appears as a valid argument form Modus Tollens (MT)

1. P > q
2. Q
3. P

MT holds because P>q is logically equivalent to (pvq) by the rule of implication (Impl) (pvq) are the conditions under which a conditional is true i.e. when either the antecedent is false or the consequent true. In these circumstances. Denying of the dysjuncts (q) justifies one to affirm the other distinct (p). such inference is valid by the rule of disjunctive syllogism (Ds).

INFORMAL FALLACIES

Informal fallacies cannot be identified by mere inspection of the argument form, but by analyzing the content of the argument. Informal fallacies can be divided into two namely, fallacies of

i. Relevance/irrelevance

ii. Ambiguity

These are errors in reasoning which have as their source the irrelevance of the conclusion to the premis(es). The error or mistake thus rests on the relevance of the conclusion to the premis(es) hence their description as fallacies of relevance or irrelevance depending on the explanation given with regard to the error. Examples of these fallacies are:

(a) **Appeal to the people or masses (argumentum ad populum)**

If numbers are quoted as evidence or justification for a certain position or conclusion, the fallacies is said to have been committed. For example, if someone is to argue that abortion, corruption, theft or murder is wrong because most people think that it is wrong, this would be an instance of this fallacy. The point to note is that the truth of falsity of something does not depend on people's opinions or feelings, rather other objective facts. The majority could believe that something is right or wrong yet all be mistaken. Question of truth and falsity, rightness or wrongs are not settled by conducting opinion polls, rather they are settled by appeal or consideration of relevant facts.

(b) Appeal to Authority (argumentum ad verecundiam)

This fallacy is said to have been committed when one accepts a statement view or position merely because an authority, expert, or a famous person accepts it or says he/she accepts it. The truth of falsity of a given statement cannot be proved merely by the fact that someone, even an authority says so. A statement is not made true or false by virtue of the prestige of an authority, rather, it is the citing of relevant and accurate evidence to confirm or refute the statement. The fact than an authority has made a statement cannot be itself regarded as evidence; it is the facts which the authority produces that constitute evidence. Such facts of course are quite different from a mere verbal pronouncement e.g.

Killing is bad or stealing is bad not merely because God or the law says so, rather because of the objective and cogent facts and evidence that points reasonably or necessarily to the badness or goodness, rightness or wrong's of the actions. Killing is bad for instance, presumably because of some good reasons which form the basis of God or the law holding that it is bad not merely because God or the law says so. Appeal to

authority rests on trust and confidence rested in authority (which is reasonable), but authority is fallible and its only on the basis of the evidence presented that we should accept the position of the authority not the mere fact of authority.

(c) Appeal to force (Argumentum and Baculem) or threat of force.

This fallacy is said to have been committed when assent is sought on the basis of the use of force or threat of force. It is not might that makes something right or true. Might or the threat of force is irrelevant when it comes to matters of truth or rightness of an action. For example, it would be fallacious if one reasoned that the United States of America war was right in 1990 with regard to the Iraq Kuwait dispute merely because Iraq was forced into submission. A preacher who argues that salvation is imperative because otherwise one would go to hell commits this fallacy because the threat of hell may compel one to declare salvation without having any objective reasons to justify the choice of salvation. The fear of the eternal suffering in hell would in that case be the reason for adopting salvation not the more reasonable grounds of love, peace, harmony and the general well being of the individual and specify that come as a result of salvation such that avoidance of hell is just an accident.

(d) Appeal to ignorance (Argumentum and Ignorantiam)

This fallacy is said to occur when it is argued that a certain view, opinion belief or assertion is true just because it has not been proved false, or conversely that it is false because it has not been proved true. This happens when the premises of an argument state that a certain position or view has into been proved (or disproved) while the conclusion makes a definite assertion abort the position. This sought of reasoning is treacherous because it apparently seems to follow and actually pretends to follow the justified reasoning that a certain view is true because we have considerable evidence, all of which shows that the view is true, and none of which shows that it is false. Showing that a view is true simply because there is no contrary evidence is not enough. It is imperative also to show positive evidence in favor of it. Otherwise outrageous claims for instance of the existence of mermaids may be 'proved' by this treacherous kind of reasoning which is logically unacceptable.

(e) Argument against the person (Argumentum and Hominem)

The above fallacy is said to have been committed when a rebut to the argument or position held is not directed to the basis, evidence or premis(es) upon which the position rests but rather to the person against whom the rebut is intended. The fallacy may appear in three ways:

I. Abusive

2. Circumstantial

3. You too (Tu Quoque)

Abusive

For example arguing that Fredrick Nietzsche was mistaken in his philosophical ideas on existentialism because he was a miserable immoral man who eventually died of syphilis involves committing this fallacy. The correctness or incorrectness' has got nothing to do with his morality. The ideas are correct or incorrect on their own merit.

Circumstantial

An example is this when for example a person who works for a certain brewing company says that one of their products is a very good beer of course presumably on the basis of some objective sound reasons. However, the fallacy is committed when one rebuts to the position by saying that the earlier person, could not be right because what he is saying is just by virtue of him being an employee of that brewing company. Important note to have is that focus is made on the circumstances of the person who says that a certain beer is good the person who happens by accident to belong to the brewing company. The beer is good or bad regardless of whether the person does or does not work for the relevant brewing company. To decide on whether or not the beer is actually good rests on a consideration of other factors such as taste, side effects and so on. However in this example, it is the circumstances of the person that are being considered to decide on whether his assertion is true or false. This is fallacious.

You too (TU Quoque)

In this fallacy, one tries to make his/her opponent appear to be arguing in bad faith. This proceeds by a citing of instances in the life or behavior of his opponent that seemingly contradict his (opponent's) conclusion. For example, a patient who argues that the doctor's advice that he (the patient) should stop drinking alcohol because alcohol is dangerous to him (the patient's) health inconsistent and therefore apparently acting in bad faith because the doctor himself drinks commits this fallacy. This is because the fact that the doctor drinks does not change the accuracy and appropriateness of the advice of the doctor to the patient to quit drinking. So long as the doctor gives such advice on the basis of objective medical considerations, even if, the doctor himself is an alcoholic, that does not render the doctors advice wrong unless it is shown that the doctors consideration are irrelevant, inconsistent or actually false.

(f) Appeal to pity (Argumentum and miserecordiam)

This fallacy is said to be committed when emotions of pity or mercy are invoked for purposes of achieving or ensuring assent. For example, when one reasons that a person should be exonerated from punishment for his/her wrongdoing because he is an orphan widow or widower or has gone through a certain traumatizing experience, not as mitigating facts but as sufficient grounds, then such reasoning is fallacious. The fact that one is an orphan for example does not make one not subject to punishment or blame for committing evils, just the same way being an orphan would not be excuse for committing evils.

(g) Fallacy of accident

The fallacy is said to have been committed when one reasons by applying a general rule to a particular instance which is uniquely circumstances rendering the general inapplicable. For example, when one reasons that because the constitution grants freedom of movement and association that therefore it is not right to prevent public rallies held by some individuals or group in the country, the fallacy would have been committed. This is because although the constitution may guarantee the freedom of movement and association there are certain conditions and assumptions under which such guarantee holds, conditions and assumptions which may hold in a particular instance thus rendering the application of the general provision of the constitution inappropriate. For example, association and movement with the objective of committing crime and disturbing the peace renders such constitutional guarantees inapplicable.

(h) Fallacy of converse Accident

The explanation of this fallacy is just the converse of that of the presiding. It proceeds by an inappropriate ascription of what holds for particular unique case to the general cases which do not experience the same and unique circumstances that the particular experiences that renders or justifies whatever it is that holds for the unique particular. For example, reasoning that since cigarette smoking increases the chances of suffering from cancer by predisposed or vulnerable individuals that then cigarette smoking should be stopped or banned altogether.

Hitherto, the various examples discussed are sufficient to illustrate fallacies of relevance/irrelevance. However, it is imperative to appreciate that the examples provided are not exhaustive since there are many more examples of such fallacies which may be seen in the references provided at the end of this lesson.

ii. Fallacies of Ambiguity

(a) Fallacy of Equivocation:

This fallacy arises from making a conclusion in an argument, which has an ambiguous word, or a word whose exact understanding is not clear.

E.g.

1. Men are the only rational creatures
2. No women are men
3. Therefore, no women are rational.

There is no certain or clear connection between the two propositions (1,2) i.e., actually there is 'almost' no middle term. This cannot be detected by an evaluation of the form or structure of the argument since this seems to present 'men' as the middle term although in

evaluating the actual content of the argument, 'men' has or at least seems to have been used in two senses, in the first sense meaning all humanity (men and women) and in the second sense meaning male human beings only. When reasoning proceeds in this way where a word is used with varying meanings in the same argument and a conclusion is drawn as if the word were used unequivocally, then the fallacy of equivocation is said to have been committee

(b) Fallacy of Amphibly

This fallacy is committed when a reasoning for instance contains a statement that could be understood or integrated in more than one senses and a conclusion is drawn as if the statement or phrase had a definite understanding or meaning. The worlds contained in the statement or phrase may not be ambiguous, yet the whole sentence is because of its grammatical structure. A common source of confusion in ordinary English stems for example from the use of the word 'not'. A sentence that begins as 'All … are not …" may be understood to mean (I) none are or (ii) some are not;

e.g. 'All politicians are not corrupt' may mean

1. No politician is corrupt or
2. Some politicians are not corrupt

The two possible interpretations become a source of confusion due to ambiguity and this renders it inappropriate from the logical point of view to draw any conclusion because of lack of certainty in the interpretation of the phrase which means that no conclusion should be drawn. However, it is important to note that the fallacy of amphiboli is committed when the amphibolors premise is given the interpretation, which makes it true, and a conclusion is drawn from it which makes it false.

e.g. All politicians are not corrupt so James who is a politician cannot be said to be corrupt!!

What is meant by the amphibolous proposition is that some politicians are not corrupt, but the conclusion is drawn as if the phrase means that no politician is corrupt.

The grammatical error can also result from dangling participles i.e. when a comma is not put in its proper place.

It should be noted at this point that the divisions and categorisation of fallacies is usually never rigid because the gist of understanding fallacies rests in the identification of the error in reasoning and how the error can be addressed. In this case, more than the fallacy may be committed and identified in one instance of reasoning. As the great logician De Morgan reiterated, we cannot pretend to be able to exhaustively classify the various ways in which mistakes can be made in reasoning.